39+30+25 : ~~99~~ 94 good!

Nama : Syaurul Ramdani Maruzah
NIM : 2413440029
Kelas : 1A-TNk /D4 Telekomunikasi
Tanggal : 9 April 2025
TTD :
Nama Matkul : Elektronika Digital Dasar

1. $F(A,B,C,D) = \Sigma m(0,2,9,10,12) + \Sigma d(8,13,14)$

a. Tabel kebenaran

Minterm	A	B	C	D	F
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	0
8.	1	0	0	0	d
9.	1	0	0	1	1
10.	1	0	1	0	1
11.	1	0	1	1	0
12.	1	1	0	0	1
13.	1	1	0	1	d
14.	1	1	1	0	d
15.	1	1	1	1	0

b. K-Map

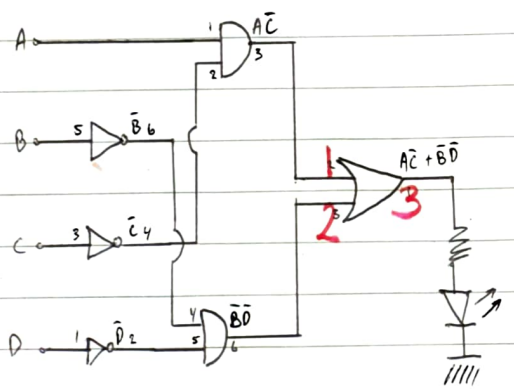
AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	1	d	0	d
10	d	1	0	1

$\bar{B}\bar{D}$

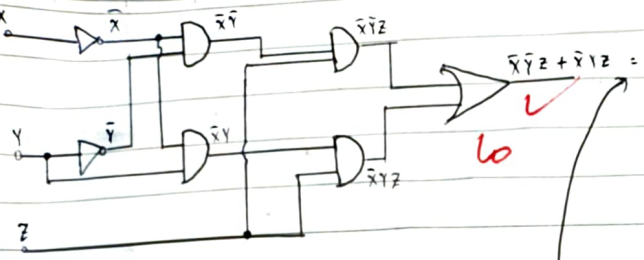
$A\bar{C}$

$F(A,B,C,D) = A\bar{C} + \bar{B}\bar{D}$

c. Rangkaian



2.



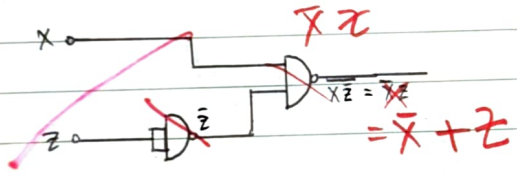
a. Tabel kebenaran

Minterm	X	Y	Z	F
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	0
7	1	1	1	0

Pers. Logika

b. $f(X,Y,Z) = \bar{X}Z$

Implementasi NAND



3. Konversi

a. $1101010,01_{(2)} = \ldots\ldots_{(10)}$

$(1\times2^6) + (1\times2^5) + (0\times2^4) + (1\times2^3) + (0\times2^2) + (1\times2^1) + (0\times2^0) + (0\times2^{-1}) + (1\times2^{-2})$

$64 + 32 + 0 + 8 + 0 + 2 + 0 + 0 + \frac{1}{4}=0,25 = 64+32+8+2+0,25 = 64+42+0,25 = 106,25_{(10)}$

b. $1C0,7_{(16)} = \ldots\ldots_{(10)}$

$(1\times16^2) + (12\times16^1) + (13\times16^0) + (7\times16^{-1}) = 256 + 192 + 13 + 0,4375 = 256+205+0,4375 = 461,4375_{(10)}$

c. $32_{(10)} = \ldots_{(16)}$

$3 = 011$, $2 = 010$ $\}$ $011010_{(2)}$ $\rightarrow$ $0001 = 1$, $1010 = 10 = A$ $\}$ $1A_{(16)}$

d. $673_{(10)} = \ldots\ldots_{(2)}$

↳ 673 : 2 = 336 Sisa 1

336 : 2 = 168 Sisa 0

168 : 2 = 84 Sisa 0

84 : 2 = 42 Sisa 0

42 : 2 = 21 Sisa 0

21 : 2 = 10 Sisa 1

10 : 2 = 5 Sisa 0

5 : 2 = 2 Sisa 1

2 : 2 = 1 Sisa 0

1 : 2 = 0 Sisa 1

$1010100001_{(2)}$

Dari bawah

e. $111000111101 01_{(2)} = \ldots\ldots_{(16)}$

↳ 0011 = 3

1000 = 8

1111 = 8+4+2+1 = 15 = F

0101 = 5

$38F5_{(16)}$